

G1 Campaign 2 — Results Acquisition Plan (Sessions R1–R4)

Adrián Lendínez Ibáñez — 30 Jul 2026. The complete test sequence to obtain the paper/thesis results, one gated batch at a time. Rule of the house: **one change per batch; no code edits during a session; a failed run is data, not a bug to fix live**. Every session starts with the connection ritual + **noisecheck** + **reLoccheck**, marker v2 running all session (w <g> records weights; heartbeat proves it alive).

SESSION R1 — Golden re-baseline (dry, ×10)

Binary: tag **golden-doorvis**, zero edits. **Physical:** cup IN THE HAND (empty), NO apron. **Hypothesis (pre-registered):** H0 = B→A success ≥70%, A→B ≥90%, times 50–110 s.

```
git fetch origin && git stash && git checkout golden-doorvis
# 10 runs (5 pairs), SAME command, alternating gotoviz B / gotoviz A:
G1_SETUP="cup=hand,apron=off" G1_ARM=REBASELINE G1_D00R_VIS=1 G1_META2=2 G1_M2_CFG=config_meta2_g1payload.json G1_M2_REALCAL=1
G1_M2_STATE=meta2_state_rebase.json G1_M2_STATE_INIT=meta2_state_twin_explored.json G1_M2_L3=1 G1_M2_L4=1 G1_M2_PROFILES=1
G1_M2_D00RLIB=1 G1_M2_FRAGSPEED=1 G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B
```

Stop rule: 3 consecutive same-mode failures → stop, push, analyse offline. **Yields:** the true baseline success rate with n; the reference for every later comparison.

GATE 1: H0 holds → proceed to R2 the same day if time allows. H0 fails → the difference is physical/environmental; diagnose from the 10 manifests, do NOT patch.

SESSION R2 — Water at 200 g (the core dataset)

Binary: still **golden-doorvis**. **Water protocol:** fill 200 g; w 200 in the marker before leaving, w <result> on arrival; never refill between the two legs of a pair; slosh burst = ONE mark; governed HELP stop = valid result. **Pair order (ABBA):** GOV, BASE, BASE, GOV, GOV, BASE.

```
# GOVERNED pair (C2-GOV) – keep the same state file all session:
G1_SETUP="cup=hand,apron=off" G1_ARM=C2-GOV G1_FILL_G=200 G1_CUP_G=246 G1_D00R_VIS=1 G1_META2=2
G1_M2_CFG=config_meta2_g1payload.json G1_M2_REALCAL=1 G1_M2_STATE=meta2_state_c2b_m2.json
G1_M2_STATE_INIT=meta2_state_twin_explored.json G1_M2_L3=1 G1_M2_L4=1 G1_M2_PROFILES=1 G1_M2_D00RLIB=1 G1_M2_FRAGSPEED=1
G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B

# BASELINE pair (C2-BASE):
G1_SETUP="cup=hand,apron=off" G1_ARM=C2-BASE G1_FILL_G=200 G1_CUP_G=246 G1_META2=0 G1_PERC=127.0.0.1:8008 python3 g1_goto.py
gotoviz A_o_B
```

Batch	n	Yields for the paper
C2-GOV 200 g	6 (3 pairs)	governed completion rate, time, marks/leg, g/mark — the headline table row
C2-BASE 200 g	6 (3 pairs)	baseline row at identical fill/code platform; paired stats (ABBA kills drift)

GATE 2: ≥4 governed legs completed → R3. Governed HELP-aborts are still data (refusal semantics) but if >50% abort, stop and analyse.

SESSION R3 — The headline (247 g) + RETREAT v2 promotion

R3a — FULLCUP 247 g (2 governed + 2 baseline legs)

Same commands as R2 with **G1_FILL_G=247**. **Yields THE claim:** campaign-1 governed aborted 4/4 at full cup (ungoverned start turns); the governed turn cap should now survive it. Baseline at 247 g closes the 2x2 fill design.

R3b — Promotion batch: main (with RETREAT v2) vs golden

The ONE allowed binary change, as its own batch. RETREAT v2 is twin-validated (6/6 clean no false fires; deterministic trap: 3 retreats undoing 0.68/0.82/0.37 m along the entry path, P9 paused during, correct abort after budget). Dry A/B: 4 runs on **golden-doorvis** + 4 runs on **main**, alternating, same physical setup.

```
git checkout main && git pull --rebase origin main      # batch B runs
git checkout golden-doorvis                          # batch A runs
# same REBASELINE command as R1 for both (G1_ARM=PROMO-GOLD / PROMO-MAIN)
```

GATE 3: main ≥ golden on success rate and times → main becomes the new working binary (tag it **golden-v2**). Otherwise stay on golden and diagnose offline.

SESSION R4 — Recovery figure + lid condition

R4a — Wrong-prior recovery (×6)

```
# fresh state ONCE: rm -f meta2_state_c2_wrong.json
G1_SETUP="cup=hand,apron=off" G1_ARM=C2-WRONG G1_FILL_G=200 G1_CUP_G=246 G1_D00R_VIS=1 G1_META2=2
G1_M2_CFG=config_meta2_g1door.json G1_M2_REALCAL=1 G1_M2_STATE=meta2_state_c2_wrong.json G1_PERC=127.0.0.1:8008 python3
g1_goto.py gotoviz B
# WRITE DOWN PL after EVERY run:
python3 -c "import json;print(json.load(open('meta2_state_c2_wrong.json')))"
```

Yields: the DST trust-erosion sequence under a wrong prior on hardware (thesis §5.3.6.3 analogue) — the recovery figure.

R4b — Sealed lid (x4: 2 baseline + 2 governed)

Same commands with the lid SEALED, `G1_ARM=C2-LID`, governed state `meta2_state_c2_lid.json`. **Yields:** covered-delivery condition + the particle-filter validation on real data (logged `meta2_pf`: P(lid open) must fall toward 0 — its first hardware ground truth).

After every block (both machines stay in sync)

```
git add dataset/ runs_summary.csv meta2_state_*.json goto.log
git commit -m "R2 C2-GOV pair 1: 200g, weights 200->161->148"
git pull --rebase origin main # (or the tag branch) -> git push
```

...and report weights + marks per leg in chat for live accounting.

Results checklist (what must exist when R1–R4 are done)

#	Deliverable	Source
1	Baseline success/time table with n=10 + H0 verdict	R1 manifests
2	Governed vs baseline at 200 g: completion, time, marks, g/mark, collisions (paired)	R2
3	FULLCUP 247 g contrast vs campaign-1 4/4 aborts (the headline)	R3a
4	RETREAT v2 real-world promotion verdict (golden vs main A/B)	R3b
5	PI erosion sequence (6 points) under wrong prior	R4a
6	Lid condition rows + PF P(lid open) validation curve	R4b
7	Photo catalog (8 shots + perception /debug.jpg) and 2–3 video clips	any session
8	Per-run manifests (env_g1 + git SHA + G1_SETUP + payload grams) — already automatic	all

Estimated lab time: R1 ≈ 60–90 min · R2 ≈ 90 min · R3 ≈ 90 min · R4 ≈ 90 min. Priorities if time is short, in this order: R1 → R2 → R3a → R4a → R3b → R4b.